Master: Amylase Rate, Graph and Topic 1 Consolidation

Today I can… ☐ Show your working: relative rate = 1 ÷ time. Use the unit s^{-1}. ☐ Read or plot how pH links to rate. Say only what the data supports. ☐ Name a limit of the method. Link it to a way to improve it. Then show how digestion differs from diffusion.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Arrival · Supported

1. Time and rate

With the same starting starch and endpoint, which is faster: 90 s or 180 s?

Shorter time for the same measured change means faster breakdown.

2. Read the endpoint

Link to today: Blue-black means starch detected. Orange-brown means starch not detected. Which colour meets our endpoint?

Endpoint = first sampled time with starch not detected.

3. Show the working

Relative rate = 1 ÷ time. Use a calculator to find 1 ÷ 100.

Press 1 ÷ 100 =. Use time in seconds; rate unit is s^{-1}.

4. Match the instruction

DESCRIBE means state the pattern. EVALUATE means judge the evidence. Which asks for a strength and a limitation?

The command word tells you what the answer must do.

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Exit Ticket · Supported

Evidence

Illustrative matched-condition data: pH 6 endpoint 120 s; pH 7 endpoint 90 s. A separate pH 7 repeat gave 120 s. Relative rate = 1 ÷ time in seconds (s^{-1}). Samples were 30 s apart; pH 4–8 were tested.

1. Calculate the pH 6 relative rate, then choose the faster first-series result: pH 6 or pH 7.

Use 1 ÷ 120. For matched endpoints, shorter time means faster breakdown.

2. Choose the stronger conclusion: “pH 7 was fastest in this first series” or “pH 7 is best for every amylase”. Give one clue.

Consider what was tested and the separate repeat.

Under the heading Exit ticket: answer the short questions in your book.